



End of Week Quiz



5/5 questions correct

Quiz passed!

[Continue Course \(/learn/advanced-data-structures/supplement/q2TbC/week-5-introduction\)](/learn/advanced-data-structures/supplement/q2TbC/week-5-introduction)

[Back to Week 4 \(/learn/advanced-data-structures/home/week/4\)](/learn/advanced-data-structures/home/week/4)



1.

What is the tightest bound Big-O running time of the brute force solution to the traveling salesperson problem (TSP), where n is the number of nodes in the graph?

☒ $O(n!)$

Well done!

You can find the explanation for why in Core: TSP Algorithm Running Time

☐ $O(n^2)$

☐ $O(2^n)$

☐ $O(n^k)$



2.

If you know that a problem is considered NP Hard, that means:

- ☐ You can be sure that the problem is at least as hard as a set of problems where there is no *known* polynomial - $O(n^k)$ - solution.

Well done!

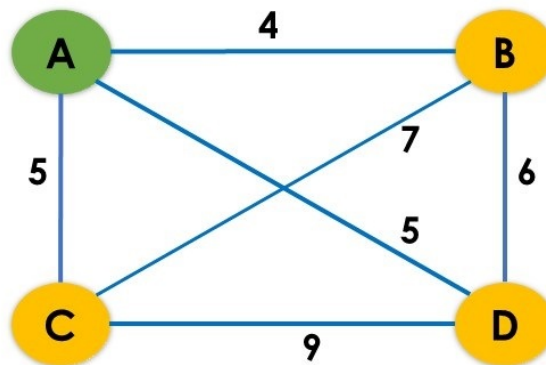
Great response! That's exactly what it means.

- ☐ You can be sure there is a polynomial time - $O(n^k)$ - solution.
- ☐ You can be sure there is no polynomial time - $O(n^k)$ - solution.



3.

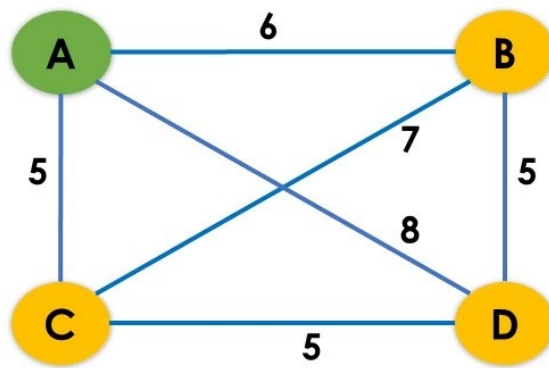
For which of the graphs below would the greedy algorithm (nearest neighbor) NOT find the best solution to the Travelling Salesperson Problem assuming A is the start vertex. Again, each node must be visited once and you must return back to A.



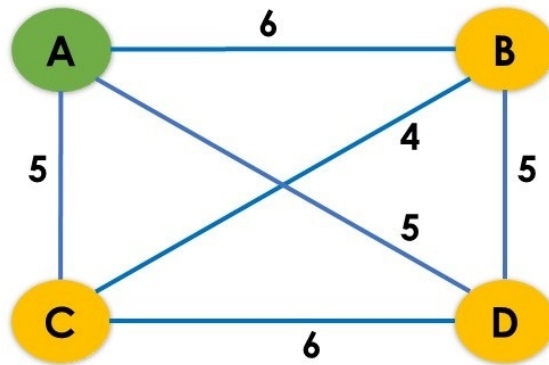
Well done!

Nicely done! Greedy would find the non optimal path of A->B->D->C->A when A->C->B->D->A is better.





☐



4.

Which of the following is true about the 2-opt heuristic as explained this week?

☐

2-opt is an approach which may improve a non-optimal solution to the TSP problem but may end up producing a solution worse than the original non-optimal solution

- ☐ 2-opt is an approach which is guaranteed to improve a non-optimal solution to the TSP problem to create an optimal solution for the TSP problem
- ☐ 2-opt is an approach which may improve a non-optimal solution to the TSP problem and is guaranteed to do no worse than the original non-optimal solution

Well done!



5.

A Hamiltonian graph is one where there's *some* path through the graph which visits every vertex exactly once. An Eulerian graph is one where there's *some* path through the graph which traverses every edge exactly once.

Which of the following is true about Hamiltonian and Eulerian graphs?

(Select all that apply.)

- ☐ Every Hamiltonian graph is Eulerian, because every Hamiltonian path uses each edge at most once.

Well done!

See the examples in the Core: Eulerian video.

- ☐ Each graph is either Hamiltonian or Eulerian.

Well done!

See the examples in the Core: Eulerian video.

- ☐ There can be a graph that is Hamiltonian but not Eulerian.

Well done!

See the examples in the Core: Eulerian video.

